

# Zhiyuan Li

Postdoctoral Researcher

Aalto Robot Learning  
Aalto University  
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## Research Interests

Multi-Agent Systems, Reinforcement Learning, Robotics, Foundation Models

## Education and Work Experience

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| <b>Aalto Robot Learning, Aalto University</b><br><i>Postdoctoral Researcher</i>                                    | Aug 2024 – Present<br>Espoo, Finland  |
| <b>University of Electronic Science and Technology of China</b><br><i>Ph.D. in Computer Science and Technology</i> | Sep 2019 – July 2024<br>Chengdu China |
| <b>North China Electric Power University</b><br><i>Bachelor in Computer Science and Technology</i>                 | Sep 2015 – July 2019<br>Beijing China |

## Publications

### **Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models**

Zhiyuan Li, Wenyan Yang, Pekka Marttinen, and Joni Pajarinen.  
[NeurIPS'26] *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026.

### **Sparsely Supervised Diffusion**

Wenshuai Zhao, Zhiyuan Li, Yi Zhao, Mohammad Hassan Vali, Martin Trapp, Joni Pajarinen, Juho Kannala, and Arno Solin.  
[NeurIPS'26] *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026.

### **Rethinking Temporal Consistency in Video Object-Centric Learning: From Prediction to Correspondence**

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Pekka Marttinen, and Joni Pajarinen.  
[ICML'26] *The Forty-Third International Conference on Machine Learning*, 2026.

### **Cycle Consistency in Video Object-Centric Learning**

Rongzhen Zhao, Zhiyuan Li, Ruonan Wei, Juho Kannala, and Joni Pajarinen.  
[BMVC'26] *The 37th British Machine Vision Conference*, 2026.

### **Learning Progress Driven Multi-Agent Curriculum**

Wenshuai Zhao, Zhiyuan Li, and Joni Pajarinen.  
[ICML'25] *The Forty-Second International Conference on Machine Learning*, 2025.

### **AgentMixer: Multi-Agent Correlated Policy Factorization**

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.  
[AAAI'25] *Thirty-Ninth AAAI Conference on Artificial Intelligence*, 2025.

### **Optimistic Multi-Agent Policy Gradient**

Wenshuai Zhao, Yi Zhao, Zhiyuan Li, Juho Kannala, and Joni Pajarinen.  
[ICML'24] *The Forty-first International Conference on Machine Learning*, 2024.

### **Coordination as Inference in Multi-Agent Reinforcement Learning**

Zhiyuan Li, Lijun Wu, Kaile Su, Wei Wu, Yulin Jing, Tong Wu, Weiwei Duan, Xiaofeng Yue, Xiyi Tong, and Yizhou Han.  
[Neural Networks'24] *Neural Networks*, 2024.

### **Backpropagation Through Agents**

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.  
[AAAI'24] *Thirty-Eighth AAAI Conference on Artificial Intelligence*, 2024.

### **Hierarchical Fusion Framework for Multimodal Dialogue Response Generation**

Qi Deng, Lijun Wu, Kaile Su, Wei Wu, Zhiyuan Li, and Weiwei Duan.

[IJCNN'24] *International Joint Conference on Neural Networks*, 2024.

### **Online Coordinated NFV Resource Allocation via Novel Machine Learning Techniques**

Zhiyuan Li, Lijun Wu, Xiangyun Zeng, Xiaofeng Yue, Yulin Jing, Wei Wu, and Kaile Su.

[TNSM'22] *IEEE Transactions on Network and Service Management*, 2022.

### **Closed-Loop Vision-Language Planning for Multi-Agent Coordination**

Zhiyuan Li, Wenshuai Zhao, and Joni Pajarinen.

[SE@AAMAS'26] *The 25th International Conference on Autonomous Agents and Multiagent Systems*, 2026.

### **Adaptive graph attention networks with interactive learning for attributed graph clustering**

Weiwei Duan, Luping Ji, Lijun Wu, Qi Deng, and Zhiyuan Li

[EAAI'26] *Engineering Applications of Artificial Intelligence*, 2025

### **Multi-agent neighborhood coordinated and holistic optimized actor-critic framework for adaptive traffic signal control**

Qi Deng, Lijun Wu, Zhiyuan Li, Kaile Su, Wei Wu, and Weiwei Duan.

[Applied Intelligence'25] *The International Journal of Research on Intelligent Systems for Real Life Complex Problems*, 2025

### **Bridging the Embodiment Gap: Disentangled Cross-Embodiment Video Editing**

Zhiyuan Li, Wenyan Yang, Wenshuai Zhao, Yue Ma, Yuanpeng Tu, Pekka Marttinen, and Joni Pajarinen.

arXiv preprint, arXiv:2605.03637, 2026.

### **Do We Really Need the Default Recipe for Video Object-Centric Learning?**

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Arno Solin, Juho Kannala, Pekka Marttinen, and Joni Pajarinen.

[TPAMI] *IEEE Transactions on Pattern Analysis and Machine Intelligence*. (Under Review)

## **Teaching Experience**

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Delivered the advanced part of Lecture 6 (Actor–Critic) in ELEC-E8125 Reinforcement Learning at Aalto University in 2026.

Main teaching assistant for ELEC-E8125 Reinforcement Learning, Department of Electrical Engineering and Automation, Aalto University, Finland (243 participants); coordinated the 12-person TA team, exercises, student project support, technical Q&A, and grading.

## **Research Funding**

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Scholarship under China Scholarship Council, Ministry of Education of the P. R. China, EUR 16,200, 2022.

Participated in grant application preparation for the 2026 International Collaboration in High-Performance Computing call of the Research Council of Finland, with Principal Investigators Joni Pajarinen (Aalto University) and Ville Hautamaki (University of Eastern Finland).

## **Awards & Honors**

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| Academic Scholarship | 2019-2022 |
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*University of Electronic Science and Technology of China*

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| Honorable Mention of Mathematical Contest in Modeling | 2017 |
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## **Community Activity**

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Area Chair for ICLR 2027.

Conference reviewer for AAAI, ICML, NeurIPS, ICLR, ECCV, and AAMAS.

## **Supervision**

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Mentored and collaborated closely with Ph.D. researchers Wenshuai Zhao (graduated December 2025) and Yi Zhao (graduated October 2026).